United International University

School of Science and Engineering

Final Examination; Year 2025; Trimester: Spring

Course: PHY 105/2105; Title: Physics; Sec: All

Full Marks: 40, Time: 2 Hours

Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.

Questions no 1, 2, 3,4 are mandatory to answer. Answer any one from question no 5 and 6.

1. (a) Two positive charges of same magnitude are placed at a distance. Why any test charge will experience no force at the middle of the two charges? Draw appropriate diagram with electric field lines. 2 CO1

(b) A dipole is placed in an electric field such that the dipole moment becomes perpendicular to the direction of electric field. Why it will experience torque? Draw appropriate diagram. 2 CO1

(c) Robert told Eduard that a ball has a charge 2×10^{-16} C while Eduard replied that the charge is not quantized. Who is correct and why? 2 CO1

2. (a) Charges of magnitude 100 μC each are located in vacuum at the corners A, B and C of an equilateral triangle measuring 4 meters on each side. If the charge at A and C are positive and the charge B negative, what is the magnitude and direction of the total force on the charge at B? 3 CO3

(b) Positive charges of 2 μC and 8 μC are placed 15cm apart. At what distance from the smaller charge will be the Coulomb force due to them be zero? Suppose the test charge inserted in between them is $+q_0$. 3 CO3

(c) Consider the two protons are separated at a distance 5 nm from each other. Compare the electrostatic force and gravitational force between them. The gravitational constant is $G = 6.6743\times10^{-11} m^3kg^{-1}s^{-2}$ and the mass of each proton is 1.67262192×10^{-27} kg. 2 CO3

3. (a) Calculate the electric field at point P from figure 01. 3 CO3

Here, $q_1 = q_3 = +5\mu C$, $q_2 = -8\mu C$, $r = 2$ cm and $R = 3$ cm.

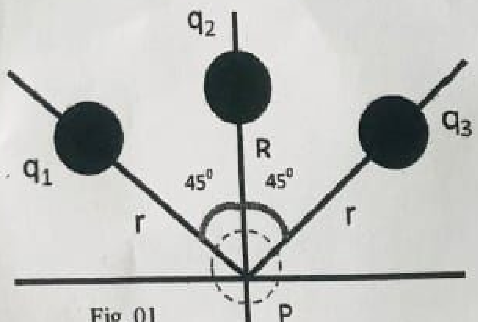

Fig. 01

(b) A dipole has charge $q = 5$ μC and the distance of the charges in the dipole is $d = 10$ nm. 3 CO3

(i) If it is placed in an electric field with value $E = 100$ N/C and the direction of dipole moment is perpendicular to the electric field, calculate the torque acting on the dipole.

(ii) How much work must an external agent do to rotate the dipole from 40^0 to 110^0 in the electric field?

(c) Calculate the electric field at point P from figure 02. Here, $q_1=q_2= 5\mu C$; $q_3= -8\mu C$ and a = 2 mm. 3 CO3

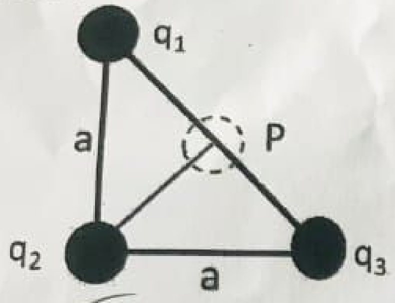

Fig. 02

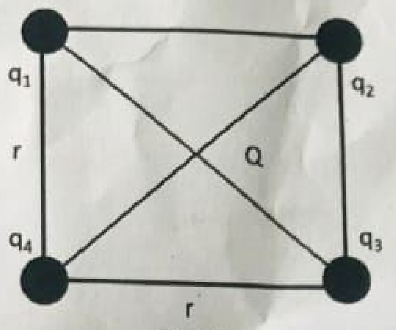

Fig. 03

4. (a) Figure 03 shows 4 charges on the four corner of a square. Here, $q_1=+5C$, $q_2= +7C$, $q_3= -8$ C, $q_4= -10$ C and r = 2 m. Calculate the net electric potential at the center of the square (at point Q). 3 CO3

(b) The ammonia molecule NH_3 has a permanent electric dipole moment equal to 1.47 D, where 1D= 1 Debye unit = 3.34×10^{-30} Cm. Calculate the electric potential due to an ammonia molecule at a point 48 nm away along the axis of the dipole. 3 CO3

(c) A charge of -2 μC is located on the y-axis at the coordinates (0,1) while a second charge of +4 μC is located on the x-axis at the coordinates (3,0) from the origin. Determine the value of electric potential at the origin. 3

5. (a) Show that, the electric field due to a dipole is, 4 CO2

$$E = \frac{1}{2\pi\varepsilon_0}\frac{p}{z^3}$$

where the symbols have their usual meanings.

(b) Show that, the potential energy of dipole is, $U = -\vec{p}.\vec{E}$ 4 CO2
Draw the diagram and discuss the conditions of maximum potential energy.

6. (a) Draw proper diagram and show that the potential difference, CO2 4

$$V_f - V_i = -\int_i^f \vec{E}.\vec{d}$$

where the symbols have their usual meaning.

(b) Show that, the electric potential due to an electric dipole, 4 CO2

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$$

Draw proper diagram and the symbols have their usual meaning.

CO1: Define different physical quantities with examples. **CO2:** Derive/Show the various equations of electric field, electric potential, electric dipole, dipole moment, electrostatic force, etc. **CO3:** Evaluate different numerical problems based on the basic characteristics of electric charge, electric field, electric potential, electric dipole moment, and electrostatic force, etc.